Algorithm 1 Multimodal Preprocessing for FRCL-GDT

Require: Raw Text String S, Raw Image File I, Max Length N, Patch Count P, Hidden Dim D

Ensure: Text Embeddings $\mathbf{T_{Emb}}$ [N $\times$ D], Visual Embeddings $\mathbf{V_{Emb}}$ [P $\times$ D], Attention Mask $\mathbf{M}_{Attn}$ [N]

```
S_clean ← Clean(S)    ▷ Remove URLs, mentions, lowercase
S_tokens ← TokenizeBPE(S_clean)
S_final ← [[CLS], S_tokens, [SEP]]
T_IDs ← ConvertToIDs(S_final)
T_Padded ← PadTruncate(T_IDs, N)
M_Attn ← GenerateAttentionMask(T_Padded)
I_norm ← ResizeNormalize(I, 224 × 224)
V_patches ← DivideIntoPatches(I_norm, P)
V_raw ← FlattenPatches(V_patches)    ▷ V_raw has shape [P × D_patch]
T_proj ← LookupEmbedding(T_Padded)    ▷ Shape [N × D]
V_proj ← LinearProjection(V_raw)    ▷ Shape [P × D]
T_PE ← LearnedPositionalEncoding(Text)
V_PE ← LearnedPositionalEncoding(Vision)
T_Emb ← T_proj + T_PE
V_Emb ← V_proj + V_PE
return T_Emb, V_Emb, M_Attn
```

Algorithm 2 FRCL Encoder and Alignment Engine

Require: $\mathbf{T_{Emb}}$ [N × D], $\mathbf{V_{Emb}}$ [P × D], $\mathbf{M}_{Attn}$ [N] (from Module I)
Require: Top-K hyperparameter K, Temperature τ, Loss Weight λ_1
Ensure: $\mathbf{H_T^{out}}$ [N × D], $\mathbf{H_V^{ranked}}$ [N × D], $\mathcal{L}_{FRCL}$

$\mathbf{H_T} \leftarrow \text{BERT\_EncoderStack}(\mathbf{T_{Emb}}, \mathbf{M}_{Attn})$
$\mathbf{H_V} \leftarrow \text{ViT\_EncoderStack}(\mathbf{V_{Emb}})$
$\mathbf{S_T} \leftarrow \mathbf{H_T} \cdot \mathbf{H_V}^\top$ ▷ Token-Patch Similarity Matrix [N × P]
for $i = 1$ to N **do**
  $S_{row} \leftarrow \mathbf{S_T}[i, \cdot]$
  $M_i^+ \leftarrow \text{TopK\_Indices}(S_{row}, K)$
  $M_i^- \leftarrow \text{NegativeSampleIndices}(S_{row}, M_i^+)$
end for
$\mathcal{L}_{FRCL} \leftarrow 0$
for $i = 1$ to N **do**
  $\mathcal{N}_i \leftarrow \sum_{j \in M_i^+} \exp(\mathbf{h}_{T_i} \cdot \mathbf{h}_{V_j} / \tau)$
  $\mathcal{D}_i \leftarrow \sum_{j' \in M_i^+ \cup M_i^-} \exp(\mathbf{h}_{T_i} \cdot \mathbf{h}_{V_{j'}} / \tau)$
  $\mathcal{L}_{FRCL} \leftarrow \mathcal{L}_{FRCL} - \log(\mathcal{N}_i / \mathcal{D}_i)$
end for
$\mathcal{L}_{FRCL} \leftarrow \mathcal{L}_{FRCL} / N$ ▷ Average loss over all N tokens
$\mathbf{H_V^{ranked}} \leftarrow \text{TensorZeroes}(N, D)$
for $i = 1$ to N **do**
  $\mathbf{H_V^{ranked}}[i] \leftarrow \text{AggregateMean}(\{\mathbf{H_V}[j] \mid j \in M_i^+\})$
end for
$\mathbf{H_T^{out}} \leftarrow \mathbf{H_T}$
return $\mathbf{H_T^{out}}, \mathbf{H_V^{ranked}}, \mathcal{L}_{FRCL}$

Algorithm 3 Smart Modality Gating Fusion

Require: $\mathbf{H_T^{out}}$ [N × D] (Contextual Text Features)
Require: $\mathbf{H_V^{ranked}}$ [N × D] (Ranked Visual Features, aligned to N tokens)
Ensure: $\mathbf{H_{fused}}$ [N × D] (Final Multimodal Feature Sequence)

$\mathbf{H}_{concat} \leftarrow \text{Concatenate}(\mathbf{H_T^{out}}, \mathbf{H_V^{ranked}})$ ▷ Sequence of vectors [N × 2D]
$\mathbf{G}_{logits} \leftarrow \text{MLP}(\mathbf{H}_{concat})$
$\alpha \leftarrow \text{Sigmoid}(\mathbf{G}_{logits})$ ▷ Generate Gating Scores [N × 1], where $\alpha_i \in [0, 1]$
$\mathbf{H_{gated}} \leftarrow \mathbf{H_V^{ranked}} \otimes \alpha$ ▷ {Visual Feature Suppression, when $\alpha_i \approx 0$, visual feature $\mathbf{h}_{V_i}$ is suppressed, filtering noise.}
$\mathbf{H_{fused}} \leftarrow \mathbf{H_T^{out}} + \mathbf{H_{gated}}$
return $\mathbf{H_{fused}}$

Algorithm 4 Module IV-A: Grid Feature Construction and Classification

Require: $\mathbf{H_{fused}}$ [N × D]
Require: Relation Tag Set R
Ensure: Prediction Grid $\mathbf{P}_{tags}$ [N × N]

$N \leftarrow \text{Length}(\mathbf{H_{fused}})$
$\mathbf{V}_{grid} \leftarrow \text{Empty\_Tensor}(N, N, 3\text{D})$
for $i = 1$ to N **do**
  for $j = i$ to N **do**
    $\mathbf{h}_i \leftarrow \mathbf{H_{fused}}[i]$
    $\mathbf{h}_j \leftarrow \mathbf{H_{fused}}[j]$
    $\mathbf{h}_{i \cdot j} \leftarrow \mathbf{h}_i \otimes \mathbf{h}_j$ {Hadamard product interaction term}
    $\mathbf{V}_{grid}[i, j] \leftarrow \text{Concatenate}(\mathbf{h}_i, \mathbf{h}_j, \mathbf{h}_{i \cdot j})$
  end for
end for
$\mathbf{P} \leftarrow \text{MLPClassifier}(\mathbf{V}_{grid})$ ▷ Output scores for all tags in R
$\mathbf{P}_{tags} \leftarrow \text{Argmax}(\mathbf{P})$ {Select the highest-scoring tag for each span (i, j)}
return $\mathbf{P}, \mathbf{P}_{tags}$

Algorithm 5 Module IV-B: Constrained DP Decoding

Require: $\mathbf{P}$ [N $\times$ N $\times$ $|R|$]
Require: $\mathbf{P}_{tags}$ [N $\times$ N]
Ensure: Final Predicted Entity Spans $\mathcal{E}$

$N \leftarrow \text{Length}(\mathbf{P}_{tags})$
$\mathcal{E} \leftarrow \emptyset$
$\mathcal{DP} \leftarrow$ Initialize_Scoring_Table(N) {DP table for optimal path scores}
for $j = 1$ to N **do**
  for $i = 1$ to j **do**
    current_score $\leftarrow$ Score($\mathbf{P}[i, j], \mathbf{P}_{tags}[i, j]$)
    if $\mathbf{P}_{tags}[i, j] \in$ {Start, Single, Middle, End} **then**
      $\mathcal{DP}[i, j] \leftarrow$ Max($\mathcal{DP}[i, j]$, current_score)
    end if
    if $\mathbf{P}_{tags}[i, j] =$ TH **then**
      for $i' = 1$ to $i - 1$ **do**
        $\mathcal{DP}[i', j] \leftarrow$ Max($\mathcal{DP}[i', j], \mathcal{DP}[i', i-1]$ + current_score) {DP update for discontinuous span using TH link}
      end for
    end if
  end for
end for
$\mathcal{E} \leftarrow$ ExtractOptimalPaths($\mathcal{DP}, \mathbf{P}_{tags}$) {Trace back paths for final spans and labels}
return $\mathcal{E}$